PR :- 5 Smart Outcome Predictor :-

PART :- A

(Q1) What are Ensemble Learning methods are why are they effective?

→ Ensemble learning combines multiple machine learning models to make better predictions than a single model. By aggregating the outputs of several learners ensembles reduce errors, improve stability, increase accuracy, and enhance model generalization.

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(Q2) Difference between Bagging, Boosting and Voting?

→ Bagging :- Trains models independently on bootstrap samples

- Reduce variance

Boosting :- Trains models sequentially by correcting previous errors.

- Reduce bias

Voting :- combines predictions of multiple models

- Improve overall accur

(Q3) What is bias- Variance Tradeoff?

→ Bias occurs when a model is too simple and underfits the data, variance occurs when a model is too complex and overfits. Ensemble methods balance bias and variance by combining multiple learners more robust predictions.

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(Q4) Explain Voting classifier and Stacking Ensemble?

→ Voting classifier :- combines predictions from multiple classifiers.

Hard voting - majority vote
Soft voting - Average probability

Stacking Ensemble :- uses predictions from base models as input to a meta-model which learns how to best combine them.

(Q5) Compare AdaBoost, Gradient Boosting, LightGBM and XGBoost?

→ AdaBoost - Reweights misclassified samples
- fast

Gradient Boosting - learns residual errors
- medium

XGBoost - Regularized Gradient Boosting
- fast

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lightGBM - leaf-wise tree growth
- very fast.